Algorithmic and High Frequency Trading

Georges Khater

March 26, 2026

Contents

1 Electronic Markets and the Limit Order Book. 1
1.1 Electronic Markets and How they Function. . . . 1
1.2 Classifying Market Participants. . . . 1
1.3 Trading in Electronic Markets . . . 1
1.4 The Limit Order Book. . . . 3

2 A Primer on the Microstructure of Financial Markets. 3
2.1 Market Making. . . . 3
2.2 Trading on an Informational Advantage . . . 5
2.3 Trading on an informational disadvantage. . . . 6

3 Empirical and Statistical Evidence: Prices and Returns 8
3.1 Introduction . . . 8
3.2 Asset Prices and Returns Intraday . . . 9
3.3 Interarrival Times . . . 9
3.4 Latency and Tick size. . . . 10
3.5 Non-Markovian Nature of Price Changes. . . . 10
3.6 Market Fragmentation. . . . 10
3.7 Empirics of Pairs trading . . . 10

4 Empirical and Statistical Evidence: Activity and Market Quality. 11
4.1 Daily Volume and Volatility. . . . 11
4.2 Intraday Activity. . . . 11

1 Electronic Markets and the Limit Order Book.

1.1 Electronic Markets and How they Function.

Review basic financial definitions.

1.2 Classifying Market Participants.

When designing Algorithmic trading, it is important to consider who trades in the exchanges and why. Since this will affect whether and how algorithms achieve the desired trading objective.

- Fundamental traders (noise / liquidity traders).
- Informed traders (active or aggressive).
- Market Makers (passive or reactive trading).

1.3 Trading in Electronic Markets

Electronic markets amount to having a way for people to show their willingness to trade with an engine to match those wanting to buy with those wanting to sell.

1.3.1 Orders and the Exchange

There are two types of orders

- Market orders; wants to buy or sell a quantity of shares at the best available price. Often results in immediate trade execution.
- Limit orders; passive trades orders which indicate willingness to buy or sell at a given price, up to a maximum quantity of shares. This usually doesn't result in an immediate trade; it will have to wait until it is matched with some other order or is withdrawn.

These trades are managed by a matching engine and a limit order book (LOB). The latter keeps track of incoming and going orders. Matching engine is a well defined algorithm to match sellers to buyers.

[insert picture] Notice how the difference in liquidity affects the LOBs.

1.3.2 Alernate Exchange Structure.

It is also important to take into consideration the degree (and cost) of transparency in an exchange. We distinguish lit (open OB) from dark markets.

1.3.3 Colocation.

Exchanges control the amount of information you receive, they also rent out computer space next to their matching engines. This process is called colocation.

We also have to keep in mind the interaction of multiple trading venues, latency when moving between these venues and regulation when designing trading strategies.

1.3.4 Extended Order Types.

The race to be the first in or out of a certain position is one of the focal points of the debate on the benefits and costs of HFT.

This importance of speed affects the whole process of designing trading algorithms (code, language, hardware, etc). Hence exchanges have moved well-beyond the basic types of orders (MOs and LOs):

- Day orders: orders for trading during regular trading with options to extend to pre- or post-market sessions.
- Non-routable: there are a number of orders that by choice or design avoid the default re-routing to other exchanges, such as 'book only', 'post only', 'midpoint peg', etc.
- Pegged, Hide-not-Slide: orders that move with the midpoint or the national best price
- Hidden: orders that do not display their quantity
- Iceberg: orders that partially display their quantity (some have options so that the visible portion will automatically be replenished when it is depleted by less than one round lot)
- Immediate-or-Cancel: orders that execute as much as possible at the best price and the rest are cancelled (such orders are not re-routed to another exchange nor do they walk the book)
- Fill-or-Kill: orders sent to be executed at the best price in their entirety or not at all;
- Good-Till-Time: orders with a fixed lifetime built into them so that they will be cancelled if not executed by its expiration time
- Discretionary: orders display one price (the limit price) but may be executed at more aggressive (hidden) prices;

Note that we should be aware of the different types of orders not only in our exchange, but across different exchanges.

1.3.5 Exchange Fee

Trading in an exchange is not free; usually a much higher fee on MOs, sometimes LOs are due rebates, or the reverse in some markets. This distorts observed market prices.

1.4 The Limit Order Book.

$$\text{Quoted Spread}_t = P_t^a - P_t^b$$

if the spread is zero the market becomes locked.

$$\text{Midprice}_t = \frac{1}{2}(P_t^a + P_t^b)$$

usually used to proxy the price of teh asset.

[insert picture] Note the difference in the LOB of liquid vs illiquid asset.

When a sell MO executes against a buy LO, it is said to *hit the bid*; otoh when a buy MO executes against a sell LO it is said to *lift the offer*.

$$\text{Microprice}_t = \frac{V_t^b}{V_t^b + V_t^a} P_t^a + \frac{V_t^a}{V_t^b + V_t^a} P_t^b$$

Where V_t^b and V_t^a are the volumes posted at the best bid and asks.

2 A Primer on the Microstructure of Financial Markets.

The goal is to consider the economics that drive these trading strategies.

For the economics of trading we look to market microstructure, as it is the subfield of finance which focuses on how trading takes place in very specific setting.

A key dimension of the trading and price setting process is that of information. Who has what information, how does that information affect trading strategies, and how do those trading strategies affect trading outcomes in general, and asset prices in particular.

What differentiates microstructure studies from more general asset pricing ones is that they focus on two aspects that are key to trading: liquidity and price discovery.

2.1 Market Making.

The market makers provide liquidity in exchange for spread, we will consider that there are a lot of market makers.

2.1.1 Grossman-Miller Market Making Model

The main risk of the above is that when accepting one side of a trade, the MM will be exposed for an undetermined period of time. Grossman & Miller (1988) provide a model that captures this problem and describes how MMs obtain a liquidity premium from liquidity traders that exactly compensates MMs for the price risk of holding an inventory of the asset until they can unload it later to another liquidity trader.

Read the section in the book describing the model, nice stuff.

The size of the liquidity shock, risk aversion, and volatility all increase the discount, while competition reduces it.

2.1.2 Trading Costs.

We saw above how the costs of holding assets affect liquidity via the costs of trading. But what drives n (the number of MMs)? We do this by linking n with participation costs. This decision to participate is determined as a function of a participation cost parameter c.

We are also interested in the effect of a cost of trading that depends on the level of activity in the market. We introduce trading costs that depend on the quantity traded, like actual exchange trading fees. By introducing such a parameter η assuming that all $|i|$ is big enough for trades to be desirable we get that the liquidity discount is bigger.

Liquidity traders incur most of the trading fees ($(1 + 2\frac{n}{n+1})\eta$), and receives less immediacy from the market.

Therefore an increase in trading has a smaller effect on liquidity via competition but a larger effect on immediacy and liquidity discount.

2.1.3 Measuring Liquidity

We now consider how these divergences from efficient prices may be observed in electronic exchanges. Note that we should (and it is easy to) adapt the analysis above to the case of an electronic market.

The Grossman-Miller model corresponds to the following sequence: as LT1's MOs enter the market, they execute against the MM's LOs in the LOB which adjusts to the incoming MO.

As the execution price changes, so does the LT1's strategy who eventually stops trading after having sold $i\frac{n}{n+1}$ shares. S_1 in this case is the average price of the executed trades.

We can write S_1 as a linear function of the quantity traded q^{LT1}:

$$S_1 = \mu + \lambda q^{LT1}$$

so that in the Grossman-Miller model we would have

$$\lambda = -\frac{1}{n}\gamma\sigma^2, \quad q^{LT1} = i\frac{n}{n+1}$$

λ here captures the market's price impact. In particular, λ is used as a proxy for liquidity of the market. A more liquid market will have a lower λ, either because of greater competition (n), lower risk tolerance γ or lower volatility σ^2.

Another way to measure liquidity based on price changes is the autocovariance of the asset's returns. Let $\Delta_1 = S_1 - S_0$ and $\Delta_2 = S_2 - S_1$, then the autocovariance of price changes is given by the following relation:

$$\begin{aligned}\operatorname{cov}[\Delta_1, \Delta_2] &= \operatorname{Cov}[\mu_1 + \lambda q^{LT1} - \mu_0, \mu_2 - \mu_1 - \lambda q^{LT1}] \\ &= -\lambda^2 \operatorname{Var}[q^{LT1}] < 0\end{aligned}$$

2.1.4 Market Making using Limit Orders.

We proposed above that MMs would participate through posting LOs, we know explain why this is a natural assumption to make. Consider a small, risk-neutral trader with costless inventory and infinite patience; she makes profit by optimally providing liquidity in an uncertain environment populated by other MMs who do not react to our MM's decisions.

Uncertainty in this context comes from the timing and size of large incoming MOs, and there are no information problems: all information is public so that everyone agrees what the current value of the asset is, which we denote by St and refer to as the midprice.

We wat to choose the distance from the midprice (the depths $\delta^{\pm}$) that maximize profit. The probability that once an MO arrives it will walk up the book to a specific price is described by $P_{\pm}$. i.e. the probability that the buy LO will be filled is $p_- P_-(\delta^-)$. We assume that the distribution of other LOs in the LOB is described by an exponential distribution with parameter κ^-. This gives us that

$$p_- P_-(\delta-) = p_- e^{-\kappa^- \delta^-}$$

Using Π to denote the MM's profit per trade, we get that the MM's optimization problem is given by

$$\max_{\delta^+,\delta^-} \mathbb{E}[\Pi(\delta^+, \delta^-)] = \max_{\delta^+,\delta^-} \left\{p^+ e^{-\kappa^+\delta^+}\delta^+ + p^- e^{-\kappa^-\delta^-}\delta^-\right\}$$

Which gives the solutions

$$\delta^{\pm} = \frac{1}{\kappa^{\pm}}$$

Note that this model is very unrealistic, the functional form of all the stochastic components of the model ($P_{\pm}$ and $p^{\pm}$) are constant and exogenous. The MM's decisions and that are other

trade ($P(\delta)$) are independent, the MM's objective function is static and very simple.

2.2 Trading on an Informational Advantage

So far we have ignored one of the main issues in trading: informational differences. Many trades happen because one of the parties has better information about what the price of the asset will be.

Informally, the model structure is:

(i) The insider observes v

(ii) After observing v, they choose $x(v)$

(iii) u is realized

(iv) The MMs observe the net order flow $x(v) + u$

(v) They compete to set $S := S(x + u)$

By semi-strong efficiency, we have that

$$S(x + u) = \mathbb{E}[v \mid x + u]$$

We hypothesize

$$S(x + u) = \mu + \lambda(x + u)$$

The insider's problem becomes

$$\max_x \mathbb{E}[x(v - S(x + u))]$$

Solving this we get

$$x^*(v) = \frac{v - \mu}{2\lambda} = \beta(v - \mu), \quad \beta = \frac{1}{2\lambda}$$

We get that

$$S = \mu + 2(x + u)\frac{\sigma_u}{\sigma}$$

So that $\lambda = 2\frac{\sigma_u}{\sigma}$; this matches the hypothesis.

2.3 Trading on an informational disadvantage.

We now put the MM at the center of the problem of trading with counterparties who have superior information.

The situation is as before, there are

- Liquidity traders.
- Informed traders.
- Competitive MMs

The MM is risk-neutral and has no cost of holding inventory.

Our model has a future cash value of the asset equal to v which we limit to two possible values: $V_H > V_L$. Let p denote the unconditional probability of $v = V_H$. All orders are of one

unit, MMs post a LO to sell one unit at price a and a buy LO at price b. Liquidity traders are price insensitive and want to buy with probability/sell $1/2$. There are many informed traders which all know v but are limited to trade a single unit: when $v = V_H$ hte buy one unit if $a < V_H$ and do nothing otherwise, similarly in the reverse case. The total population of liquidity / informed traders is normalized to one, α are informed and $1 - \alpha$ are uninformed liquidity traders.

The MM's problem is to choose a and b in this setting. Competition between MMs drives their expected profits to zero. Rather than solve for a and b directly, define the ask/bid half-spreads Δ_a and Δ_b resp. The sum of the two $\Delta_a + \Delta_b$ represents the quoted spread. Let

$$\mu = \mathbb{E}[v, \mathcal{F}]$$

Then MMs chose

$$a = \mu + \Delta_a, \; b = \mu - \Delta_b$$

Consider what happens when a buy order comes in:

- uninformed liquidity trader: profit is $a - \mu = \Delta_a$.
- Informed trader: loss is $a - V_H = \Delta_a - (V_H - \mu)$

For the MM the probability that a liquidity trader wants to buy is $1/2$ and that an informed trader wants to buy is p. Therefore the expected profit is

$$\frac{(1-\alpha)/2}{\alpha p + (1-\alpha)/2}\delta_a + \frac{p\alpha}{\alpha p + (1-\alpha)/2}(\Delta_a - (V_H - \mu))$$

Setting this to zero we get

$$\Delta_a = \frac{1}{1 + \frac{1-\alpha}{\alpha}\frac{1/2}{p}}(V_H - \mu)$$

similarly

$$\Delta_b = \frac{1}{1 + \frac{1-\alpha}{\alpha}\frac{1/2}{1-p}}(\mu - V_K)$$

2.3.1 Price Dynamics.

We can extend the model above by

1. incorporating a time dimension
2. making liquidity traders price sensitive

The former is simple, in order to avoid keeping track of the interest rate, set it to zero and index all variables by time t and set the time of determination of cash value to T. Also ensure that everything incorporates the accumulation of public information from trade, as captured by $\mathfrak{F}_t$. MM's set $p_t = \mathbb{P}(v = V_H \mid \mathcal{F}_t)$ and $\mu_t = \gtrapprox \mid \mathcal{F}_{\approx}$. We get

$$a_t = \mu_t + \Delta_{a,t}$$

and

$$b_t = \mu_t - \Delta_{b,t}$$

Notice that the realized price process is a martingale.

2.3.2 Price Sensitive Liquidity Traders.

We can also extend the model so that liquidity traders avoid trading if the half spread Δ is too high. We can do this by assuming that liquidity trader gets a utility gain of c_i if he manages to execute his trade. Let $F(c)$ be the CDF that describes the distribution of c_i in the population, i.e. $F(c)$ is the proportion of traders with $c_i \leq c$; c_i is the agent's urgency parameter. We get that the expected profit from the MM by setting $a = \mu + \Delta$ is given by

$$\frac{(1 - F(\Delta_a)(1-\alpha)/2)}{\alpha p + (1 - F(\Delta_a)(1-\alpha)/2)}\Delta_a + \frac{p\alpha}{\alpha p + (1 - F(\Delta_a)(1-\alpha)/2)}(\Delta_a - (V_H\mu))$$

This parameter increases the half-spreads, which are now implicitly defined by

$$\Delta_a = \frac{1}{1 + \frac{1-\alpha}{\alpha}\frac{(1-F(\Delta_a)/2)}{p}}(V_H - \mu)$$

similarly

$$\Delta_b = \frac{1}{1 + \frac{1-\alpha}{\alpha}}\frac{(1 - F(\Delta_a)/2)p}{(}\mu - V_L)$$

The issue here is that if the urgency parameter in the population are too small; therefore the MM may find that the above expressions have only the solution $\Delta_a = V_H - \mu$ and $\Delta_b = \mu - V_L$. In this case there is a market collapse: no one gains anything from trade and any trade that may occur will come from the informed agents. This immediately reveals the underlying value of the asset and the price will be strong-efficient.

3 Empirical and Statistical Evidence: Prices and Returns

3.1 Introduction

This chapter first looks at millisecond data . We then turn to look at the properties of returns at the daily at much shorter time intervals as well as looking at the interrarival times of price changes.

3.1.1 The Data

The data are made up of the following fields:

- Timestamp
- Order Immediate
- Message Type
  - "B" - Add buy order
  - "S" - Add sell order
  - "E" - Execute outstanding order in part
  - "C" - Cancel outstanding order in part
  - "F" - Execute outstanding order in full

  - "D" - Delete outstanding order in full
  - "X" - Bulk volume for the cross event
  - "T" - Execute non-displayed order
- Shares
- Price
- Ticker
- MPID: Market participant id
- Exchange

3.1.2 Daily Asset Prices and Returns

When trading, the first variable of interest is the price level. If we have to ac quire/liquidate a position we want to know what price we can get if we aggressively execute it, and if we are providing liquidity we want to know at what prices shares are being bought and sold.

Market efficiency theories tell us that the price process is not predictable and any positive expected return you can predict, is there as compensation for bearing risk. Therefore long-term investors receive compensation for risk. Similarly Liquidity provider require compensations for leaving offers at the bid and ask.

3.1.3 Daily Trading activity.

Measures of trading activity include: Average number of transactions per day on NASDAQ (N), average total daily dollar value of shares traded on NASDAQ ($V(\$)$ in 000s), average number of shares traded daily on NASDAQ ($V(Q)$ in 000s), total average number of shares traded in all markets (Total $V(Q)$) and Share Turnover.

3.1.4 Daily Price Predictability.

According to the efficient market hypothesis, daily asset returns should be close to unpredictable and reflect information in the market. To investigate this we run ordinary least squares (OLS) regressions for intra-day (market open to market close) returns for our four assets.

A variable of importance is order flow: In electronic markets we differentiate between posted LOs and MOs. If a transaction is the result of a limit sell (buy), being lifted (hit) by an aggressive market buy (sell) we refer to it as an aggressive buy (sell) order. Notice that this is a proxy for the net demand for the asset which can incorporate information important to market making strategies (Chapter 2).

View models in the book and results in the book.

3.2 Asset Prices and Returns Intraday

The asset's return is evaluated using the Microprice S_t^* (also called the weighted-midprice); it is given by

$$S_t^* = \frac{V_t^b}{V_t^b + V_t^a} P_t^a + \frac{V_t^a}{V_t^b + V_t^a} P_t^b$$

Where $P_t^\bullet$ is the best $\bullet$ price and $V_t^\bullet$ are the relative quantities posted at $\bullet$.

Note that the choice of sampling frequency is important as it has a very significant effect on the properties of the empirical distribution of the asset's return. If it is too short then many of the observation will be zero.

We assume that the tails fo the return follow a power law with a probability distribution function given by $f(r)$ where

$$f(r) = \frac{\alpha - 1}{r_{min}} \left(\frac{r}{r_{min}} \right)^{-\alpha}$$

The MLE for α can be shown to be

$$\hat{\alpha} = 1 + T \left[\sum_{t=1}^{T} \log(\frac{r_t}{r_{min}}) \right]$$

To judge whether the market behaves in accordance with efficient market hypotheses, we estimate the ACF from the return data. The ACT $f(n)$ is given by the correlation of r_{t-n} with r_t.

3.3 Interarrival Times

Let τ_i denote the times at which there is a change either in the bid or in the ask abd we look at the frequency of interrarival times $X_i = \tau_{i+1} - \tau_i$.

3.4 Latency and Tick size.

Any agent for whom immediacy is very important needs to account of a new issue: message latency.

When executing an MO agents need to be aware of the relationship between her choice of routing strategy and how other traders may react to the information that may be extracted from observing the strategy's outcomes.

3.5 Non-Markovian Nature of Price Changes.

We look at how price changes are interrelated. By looking at the tables, there seems to be little to no difference between conditional and the unconditional transitions. Hence, even though price changes tend to be reversed, the direction of the current price change does not carry any additional info about future price changes.

3.6 Market Fragmentation.

We must also note the issue of Market Fragmentation. We cannot talk about 'the market' as a single exchange, but as the aggregation of activity across a large number of venues. Market fragmentation is the extent to which an asset is distributed across different venues.

Optimally executing a trade is nto just about timing and prices displayed in one exchange, but also about how to organize the way an order reaches a particular trading venue

3.7 Empirics of Pairs trading

Most traders do not look at one asset at a time, but consider the interactions between different assets.

Our analysis is based on the following the following model: We assume that both INTC and SMH are stocks whose dynamics have a transitory (mean-reverting) component and a permanent (Brownian) component; we express the dynamics of this process in vector form as follows:

$$dS_t = \kappa(\theta - S_t)dt + \sigma dW_t$$

Where $\Sigma = \sigma\sigma'$ and W_t is Brownian motion. By finding the eigenvalues of κ, the largest (in absolute terms) will have the strongest exposure to the mean-reverting process, and hence should contain the most trading-relevant information.

To estimate the discrete version of the model, we estimate the vector autoregressive proecess (VAR)

$$\Delta S_t = A + B\Delta S_{t-1} + \varepsilon_t$$

Doing some arithmetic this allows us to estimate $\kappa,\ \theta$. We can then diagonalize κ and get the eigenvalues and the new price coordinates in this new basis. Picking the bigger of the two (here $\tilde{S_{t,1}}$) this will be the more strongly mean-reverting one. Hence deviation of $\tilde{S_{t,1}}$ from its equilibrium are corrected more quickly and therefore are more predictable.

4 Empirical and Statistical Evidence: Activity and Market Quality.

4.1 Daily Volume and Volatility.

So far we have seen that the price level over the course of a whole da are difficult to predict and move with market forces. But trading activity has a different dynamic structure that affects the way we look at market data.

The idea is that any variable correlated with volatility will inevitably posses forecasting power for future volatility. This is true for bid-ask spreads, transaction count, normalize trading volume, ...

First we look at the relationship between volume and volatility using regression. [View their results]

volume seems to have significant time series persistence as evidenced by the common, positive and significant coefficient on last period's volume (Ml and M2 for all assets), and positive and significant 'correlation' with volatility, as measured by HL-volatility (in M2, all assets).

4.2 Intraday Activity.

We notice empirically a large peak in intraday volume at the beginning and end of the day. We also notice a large variability in volume, which confirms the impression that intraday volume has large peaks of trading.

We notice that this pattern also happens at the 30- and one-minute windows. We observe large peaks of trading mixed in with periods of relative calm.

4.2.1 Intraday Volume Patterns.

To explain the peaks at the beginning and end of the day, we need to hypothesize about the factors that drive volume. A common hypothesis is that new information generates greater volume. This would explain the large volume at and just after the opening, as overnight news is gradually incorporated into prices during that time. But this doesn't explain the magnitude of the increase in volume at the end of the day. We will later see that this might be due to traders who have not yet met their liquidations target and will accelerate trade execution as the market approaches its time to close. A second explanation might be that traders may prefer to postpone non-urgent execution towards the end of the day when execution costs are lower. Finally, any strategy that targets volume will naturally exacerbate the increase in volume anticipated during this period.

Also read about how FDA curves help identify outliers in the data. Which is important when backtesting since these can have disproportionate effect on an algorithm's profitability.